Children's narrow cognitive bottleneck accelerates the emergence of the statistical properties of language

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While language use is rich and has the potential for infinite expressivity, learners are limited in how much they can learn when a language is passed from one generation to the next. This transmission bottleneck is thought to play a crucial role in shaping language by promoting the emergence of systematic structure that enhances learnability (Brighton, 2002; Kirby, 2000; Smith, 2003). Prior experimental and computational work suggests that narrow bottlenecks accelerate the emergence of systematic structure, because learners must generalise from limited input, whereas wider bottlenecks lead to more faithful transmission, slowing down the emergence of structure (Kirby, Dowman, & Griffiths, 2007). Bottlenecks are typically imposed experimentally by varying how much of a behaviour learners are exposed to. However, cultural transmission is also constrained by limitations on learners' cognitive capacities, such as attention and memory (Kirby, Tamariz, Cornish, & Smith, 2015; Christiansen & Chater, 2016). It remains unclear whether such cognitive bottlenecks similarly influence the emergence of systematic structure.

In this study, we present a novel approach to address this question by comparing cultural transmission in two populations with differing cognitive constraints: children and adults. Prior research on emerging sign languages and artificial language learning suggests that children may introduce more systematic structure than adults, potentially due to their more limited cognitive resources (Senghas & Coppola, 2001; Clay, Pople, Hood, & Kita, 2014; Hudson Kam & Newport, 2009). However, the few studies that have directly tested how cultural transmission by children affects the emergence of systematic structure report mixed findings (Raviv & Arnon, 2018; Kempe, Gauvrit, & Forsyth, 2015; Kempe, Gauvrit, Gibson, & Jamieson, 2019), suggesting that while limited cognitive capacities promote the emergence of structure, too narrow bottlenecks can limit learning to the point where the emergence of structure is hindered.

Our study investigates how cognitive bottlenecks shape the emergence of two cross-linguistic statistical properties that were shown to facilitate learning: sta-

tistically coherent parts, where transitional probabilities of the elements within parts are higher than across boundaries (Saksida, Langus, & Nespor, 2017; Saffran, Aslin, & Newport, 1996), and Zipfian frequency distributions (Zipf, 1949; Lavi-Rotbain & Arnon, 2022). Following prior work with adults (Arnon & Kirby, 2024; Wolters, Kirby, & Arnon, 2025), chains of children and adults observed and reproduced sets of colour sequences produced by the preceding participant, simulating cultural transmission. Each participant reproduced 30 sequences composed of four colours (initial length = 10), across six chains of ten generations per age group, reproducing one sequence per trial using a Simon-game-like task.

Children made significantly more errors than adults when reproducing the initial randomly generated sets, confirming their more narrow cognitive bottleneck. However, by the end of transmission, their accuracy matched that of adults (Fig. 1). To analyse the statistical properties of the sets, sequences were segmented into parts where there were dips in the transitional probabilities of the colours, using a segmentation method inspired by infant statistical-learning research (Arnon & Kirby, 2024). For both age groups, sets evolved to have statistically coherent parts (measured as the mean transitional probability of colours within parts) with a Zipfian frequency distribution (measured as the R^2 fit to a linear log-log regression), and crucially, these properties emerged faster with children (Fig. 1).

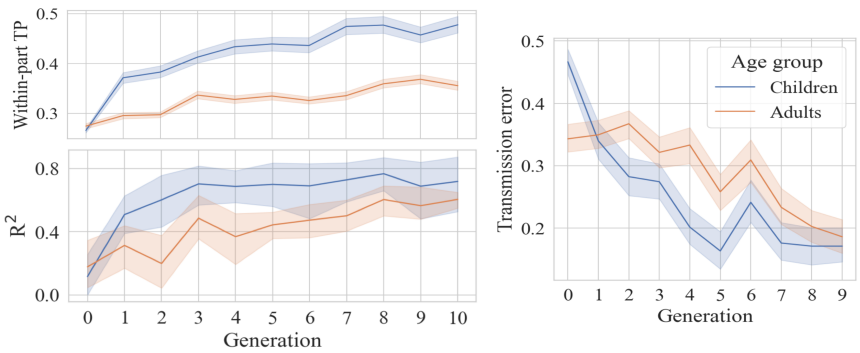

Figure 1. *Top-left*: Mean within-part transitional probability (TP) of parts over generations. *Bottom-left*: Mean R^2 values of chains over generations. *Right*: Mean transmission error of chains over generations. Error-bars represent the 95% confidence intervals.

Taken together, our findings suggest that children's more limited cognitive capacities create a narrow bottleneck that accelerates the emergence of characteristic statistical properties of language that enhance learnability. With that, we provide experimental evidence that bottlenecks resulting from cognitive constraints can modulate the emergence of structural properties that facilitate learning.

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